Cascading with Confidence: A Unified Benchmark for LLM-Powered Semantic Filter Processing [Experiment, Analysis & Benchmark]

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ABSTRACT

Semantic filters enable natural language predicates over unstructured documents, but evaluating every document with a large language model (LLM) oracle is expensive. Proxy cascades reduce oracle invocations using cheaper surrogates, but oracle reduction alone cannot distinguish proxy limitations from calibration bottlenecks or establish practical efficiency. We present a unified benchmark of four representative systems, which formulate query processing as a three-stage workflow comprising proxy construction, threshold calibration, and cascade inference, under two threshold acceptance rules. We introduce cascadability: the maximum empirical oracle reduction attainable from fixed proxy scores at a specified accuracy target, computed retrospectively using complete oracle labels. We investigate 8 research questions, spanning proxy quality, end-to-end performance, workload sensitivities, and component ablations, and reveal our key findings accordingly. Comparing cascadability with achieved oracle reduction yields a realization gap that, interpreted alongside accuracy, exposes distinct proxy and calibration bottlenecks. The benchmark and diagnostics support evaluation beyond oracle reduction and motivate joint proxy–calibration design, class-sensitive quality assessment, and end-to-end cost optimization. We release a dataset-query benchmark, containing 10 real-world workloads across various domains and difficulties, with annotations of two LLM families.

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The source code, data, and/or other artifacts have been made available at https://github.com/Anto-an/semantic_filter_benchmark.

1 INTRODUCTION

Semantic filters extend classical database selection to natural language predicates over textual attributes or unstructured documents, allowing users to express semantic conditions that are difficult to express using symbolic selection operators [44, 50]. A reference physical execution evaluates the predicate on every document using a large language model (LLM), incurring substantial query latency and monetary cost. To reduce this overhead, semantic filtering systems have proposed proxy cascades that employ cheaper surrogate models or proxy tasks to pre-filter documents, and direct unresolved documents to the reference LLM [44, 50, 51, 60]. The design objective is not simply to replace the expensive LLM with cheaper models, but to reduce execution cost under a specified accuracy quality target. The accuracy target is typically measured as the agreement with the reference LLM that serves as an oracle.

Evaluating whether a cascade achieves this objective requires distinguishing the opportunity available to the system from its ability to exploit that opportunity. The oracle reduction rate (ORR) measures the fraction of documents bypassed by the proxy, directly determining the cost savings of oracle invocations. Systems generally favor high ORR under the quality target, but ORR alone cannot cannot diagnose proxy limitations from calibration bottlenecks or establish practical efficiency. Precisely, low ORR can arise for two different reasons: the proxy itself may provide little opportunity to bypass the oracle invocations while preserving the required accuracy target, or the proxy scores may support substantial reduction, but the system fails to calibrate effective proxy thresholds to realize the reduction. Furthermore, a high ORR does not necessarily indicate an effective cascade, which may accompany violations of the accuracy target. In addition, the costs of proxy construction, proxy scoring, and threshold calibration are not negligible and may offset the savings from the reduction of oracle calls. Therefore, understanding cascade performance requires connecting available opportunity, realized reduction, and practical efficiency.

Existing systems make different design choices at each stage of proxy construction and proxy cascade. For example, *LOTUS* [44] and *BARGAIN* [59] adopt small LLMs as surrogate models but differ in threshold calibration and quality estimation. *ScaleDoc* [60] trains a lightweight surrogate on document embeddings, while *Task Cascade* [51] constructs a multi-stage cascade of task-level proxies by agentic greedy search. These differences make it difficult to attribute observed performance to the surrogate alone or to the procedure. Although prior studies have examined several aspects of quality and efficiency, the relationship between proxy opportunity and its realization remains insufficiently characterized across these designs under comparable conditions. We investigate this relationship through 8 research questions covering proxy discrimination and confidence, attainable and realized oracle reduction, target

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satisfaction and execution cost, workload sensitivity, and cross-framework component combinations. Together, these questions move the evaluation beyond identifying which system performs best to understanding where its gains and limitations arise.

We present a unified benchmark and experimental study that organizes the four representative systems into three phases: proxy construction, threshold calibration, and cascade inference. We evaluate them under an accuracy target, with explicit adaptations where their native formulations differ. The benchmarking is conducted on 10 dataset–query workloads spreading the original testing of the four systems with two LLM families. We propose a retrospective diagnostic, cascadability κ_α, for proxy cascades under the two-threshold acceptance rule, which is the maximum oracle reduction attainable from a fixed proxy score distribution at a specified accuracy target α. Comparing this quantity with achieved ORR yields a realization gap, which we interpret together with achieved accuracy to distinguish limited proxy opportunity from incomplete exploitation. We complement this analysis with target satisfaction measurements, token costs, end-to-end latency, and component-switch ablations.

Across 8 research questions, our benchmarking reveals complementary findings spanning proxy behavior and confidence, cascadeability and realization gap, reliability, efficiency, workload sensitivity, and cross-component ablations, where limitations exist in both cascadeability and realization across all the systems. Altogether, these findings move beyond system rankings to identify distinct barriers to reliable and efficient semantic filtering. In a nutshell, semantic filter systems favor joint proxy–calibration design, workload-aware and class-sensitive quality control, and end-to-end cost optimization. Finally, we outline 5 future research directions that broaden the scope of semantic filter processing. Within current scope of single-predicate filters over static document collections, the contributions of this paper are summarized as follows:

- **An empirical characterization of cascade bottlenecks and trade-offs.** We distinguish limitations in proxy opportunity from limitations in its realization, and connect these observations to confidence behavior, target satisfaction, token cost, and latency.
- **A common evaluation framework with retrospective diagnostics.** We place four representative systems within a unified framework and evaluate them by a common protocol, making their proxy-construction and calibration choices explicit.
- **A comprehensive semantic filter test suite.** We collect and release a dataset-query test suite, containing 10 real-world workloads with annotations of two LLM families. Based on these datasets, we also construct specific workloads with controlled query selectivities and oracle confidence score distributions.

The complete evaluation results on two LLM families, Qwen and Gemma, are reported in our online technical report [1].

2 PRELIMINARIES

Semantic Filters. Let $D = \{x_1, \ldots, x_n\}$ be a collection of n documents and q a natural-language predicate, such as "the opinion reverses a lower-court ruling". A semantic filter returns the documents in D that satisfy q. The *reference execution* of a semantic filter invokes a capable yet expensive oracle model, i.e., a Large Language Model (LLM), M_o on every document, and the obtained reference label of a document x_i is denoted as $y_i = M_o(x_i, q) \in \{0, 1\}$, where $y_i = 1$ indicates that M_o judges x_i to satisfy q, and $y_i = 0$ otherwise. Thereby, the reference execution invokes M_o n times to process the collection D, which is expensive w.r.t. both token cost and latency.

Proxy Cascade. Semantic filter systems reduce the cost of reference execution with a *proxy cascade* paradigm [44, 51, 59, 60]. These systems employ an inexpensive surrogate model M_s that produces a proxy score $\widehat{s_i} = M_s(x_i, q) \in [0, 1]$ for each document $x_i \in D$, indicating how likely x_i satisfies q. Throughout this paper, a proxy stage uses a *two-threshold* acceptance rule to process the documents D. The two thresholds $\tau^-, \tau^+ \in [0, 1]$ with $\tau^- \le \tau^+$ are calibrated from a small sample of documents $D' \subset D$ whose oracle labels are collected by invoking the oracle model M_o. Proxy cascade, therefore, processes remaining documents $D \setminus D'$ in a two stage paradigm:

- Surrogate invocation. If $M_s(x_i, q) \le \tau^-$ or $M_s(x_i, q) \ge \tau^+$, the system immediately returns False or True, respectively.
- Oracle fallback. If $M_s(x_i, q) \in (\tau^-, \tau^+)$, the system resorts to the oracle model M_o and returns the result $M_o(x_i, q)$.

On the calibration set D', the two thresholds τ^-, τ^+ are calibrated according to the expectation of the quality target:

$$\mathbb{P}\left[\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(\widehat{y_i} = y_i) \ge \alpha\right] \ge 1 - \delta, \tag{1}$$

where $\widehat{y_i}$ is the final output of the cascade, α is the accuracy target and δ is the failure probability. Although different systems support different quality targets such as Accuracy, Precision, Recall, and F1, without loss of generality, we use the Accuracy target in Eq. (1) as the quality target in this benchmark because it incurs minimal changes to the systems being studied.

3 A UNIFIED PROXY CASCADE FRAMEWORK

We unify the deployment of the proxy cascade in the systems being studied into a general framework for a holistic review, although there are considerable differences in specific implementations. As illustrated in Fig. 1, a workload consists of a document collection D and a natural-language predicate q. The framework comprises three phases: proxy construction, threshold calibration, and cascade inference.

Proxy Construction. This phase constructs one or more inexpensive proxies that assign a score $\hat{s}_i$ to each document x_i. Fig. 1 illustrates three representation designs: direct small-LLM, trained lightweight, and task-level proxies. System instantiations differ along five dimensions: *Surrogate model* is either a smaller LLM or a lightweight model trained per predicate on precomputed document embeddings; *Proxy score* is derived from the surrogate model and is either a token probability, the confidence of the predicted label, or a similarity between embeddings, indicating the likelihood that the document passes the predicate; *Task* is either the original predicate or a surrogate task that is easier than the oracle, such as a prerequisite predicate; *Input* is either the original document, or pruned or reordered chunks of the document, or the embedding of the document computed offline; *Supervision* involves zero-shot inference, few-shot prompting, or supervised training using oracle-labeled examples.

[1] xxx.

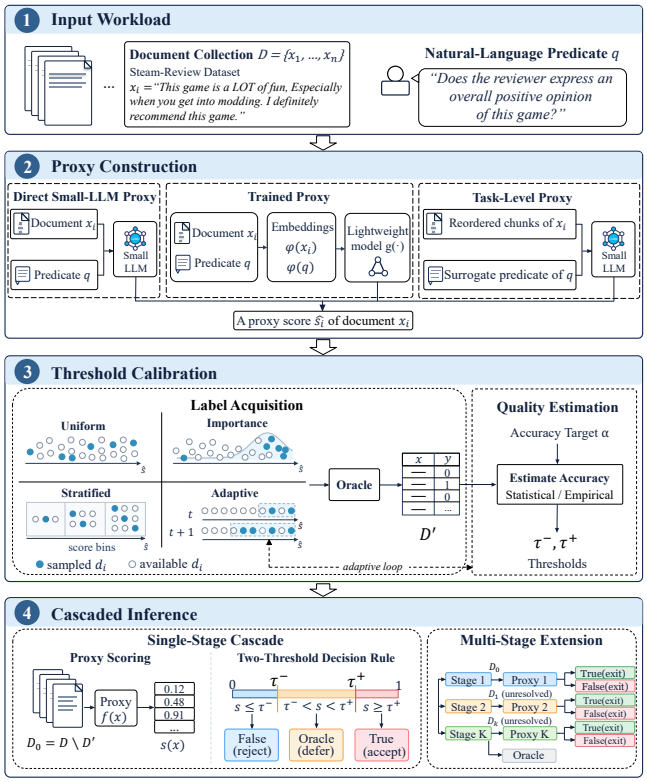


Figure 1: Cascade framework of semantic filter

Threshold Calibration. Given the proxies, this phase selects two thresholds τ^-, τ^+ from a calibration set $D' = \{x_1, \cdots\}$ where each document x_i is annotated with its oracle label y_i, following the two-threshold acceptance rule and the default Accuracy target in Eq. (1). This phase comprises two intertwined design dimensions. *Label acquisition* decides which and how many documents to label as the calibration set $D' \subseteq D$, obtaining corresponding oracle labels y_i. Systems employ uniform, importance, stratified, or adaptive sampling, potentially guided by proxy scores. Sampling may be performed in a single round or iteratively, with subsequent selections informed by previous observations and quality estimates. *Quality estimation* decides how to evaluate the Expected Accuracy and quantify the uncertainty, using empirical estimates or statistical methods such as the Central Limit Theorem, a betting-based test, or Bernstein bound. These estimates guide the selection of a pair of thresholds τ^- and τ^+ for each proxy stage. In multi-stage systems, the phases of proxy construction and threshold calibration may be interleaved, and multi-stage proxies are jointly optimized.

Cascaded Inference. With the cascaded proxy and the thresholds, the framework processes each document $x_i \in D \setminus D'$ using the two-stage workflow of surrogate invocation and oracle fallback. Multi-stage cascade extends the two-stage workflow, where each stage processes only the documents that reach the stage.

4 SYSTEM UNDER STUDY

Table 1 summarizes how each system instantiates the proxy cascade framework. For each system being studied, we describe its native design and our configuration for the common testing protocol (§4.1), and elaborate on their quality estimation and control (§4.2).

4.1 Native Design & Our Configuration

LOTUS [44]. *LOTUS* employs a pretrained smaller LLM as a surrogate model that evaluates the predicate q on every document in D. The proxy scores are the log-probabilities of the output True/False Token, normalized to the index of the quantile bucket over the whole collection (50 buckets by default). Following SUPG [28], *LOTUS* draws an importance sample with replacement. Sampling probabilities are proportional to the square root of the proxy score and are defensively mixed with uniform sampling. *LOTUS* labels the sample with the oracle. Using the central limit theorem and the normal approximation on the samples, *LOTUS* selects τ^+ and τ^- to reach user-specified Precision and Recall targets, respectively, by allotting a failure probability $\delta/2$. The native implementation of *LOTUS* does not support an Accuracy target. For unified benchmarking, we extend *LOTUS* with an Accuracy target by keeping its proxy scores, sampling design, and normal-approximation bounds, and replacing the estimated metric to select τ^-, τ^+.

BARGAIN [59]. Similar to *LOTUS*, *BARGAIN* evaluates q on each document with a smaller LLM. However, it has different variants that support Accuracy, Precision, and Recall targets under finite-sample guarantees. We use the Accuracy target variant in this paper. Taking the predictive probabilities of the output True Token directly as the proxy scores, *BARGAIN* partitions all documents into score regions defined by pre-specified intervals (e.g., 20 score subranges), where each region identifies a candidate threshold. *BARGAIN* starts sampling from the region with the highest proxy scores and progressively expands to lower-score regions. Within each region, it samples tuples and invokes the oracle LLM to test whether the tuples in that region can meet a user-specified quality target under the Betting-based test [23]. The region-wise procedure terminates once *BARGAIN* identifies a minimum threshold τ such that including any lower-score region would violate the target quality. The native implementation of *BARGAIN* is a single threshold cascade. Here, we adapt it to a two-threshold cascading framework by renormalizing the proxy's True/False token probabilities and setting $\tau^- = 1 - \tau$ and $\tau^+ = \tau$.

ScaleDoc [60]. *ScaleDoc* uses a multilayer perceptron (MLP) as the surrogate, which is trained per query on document embeddings. Given uniformly sampled training data annotated by the oracle model, the MLP is trained by contrastive learning that distinguishes the positive and negative samples and enforces bipolarity in the latent space. The cosine similarity between the resulting document embedding and the query embedding serves as the proxy score. For threshold calibration, *ScaleDoc* first partitions score ranges into bins, then draws a separate calibration sample by stratified sampling with allocation proportional to bin populations, and labels the samples by oracle invocation. From the labeled sample, it reconstructs the score density by linear interpolation and moving average smoothing. For candidate threshold pairs on the bin boundaries, *ScaleDoc* estimates the proxy quality and selects the thresholds with the lowest oracle deferral subject to the Accuracy target.

Table 1: Comparison of proxy construction and threshold calibration methods.

	Proxy Construction					Threshold Calibration	
	Surrogate model	**Proxy Score**	**Task**	**Input**	**Supervision**	**Label acquisition**	**Quality estimation**
LOTUS [44]	small LLM	Quantile-normalized predictive probability	original predicate	original document	zero-shot	Importance & Uniform sample	CLT bound
BARGAIN [59]	small LLM	predictive probability	original predicate	original document	zero-shot	Target-aware adaptive sample	Betting-based test
ScaleDoc [60]	multilayer perceptron	Cosine similarity of embeddings	original	document embedding	collected training data	Stratified sample	Bernstein bound
Task Cascade [51]	small LLM	predictive probability	original or agent-generated predicate	prefix of reordered document chunks	few-shot	Uniform sample	Empirical

Task Cascade [51]. *Task Cascade* generalizes the two-stage proxy to a multi-stage proxy, where each stage is a task-level proxy, specified by a surrogate LLM, a surrogate predicate, and a fraction of document chunks. First, each document is restructured so that the content most relevant to the original predicate q appears first. Specifically, *Task Cascade* reorders the chunks of each document by their relevance to q, and a stage (task) reads only a prefix of the chunks of its subsequent stage, with full document fallback to the oracle, which enables prefix cache reuse. Guided by a development set, an LLM agent then iteratively adds surrogate predicates, i.e., simpler, decomposed, or otherwise correlated predicates. *Task Cascade* sets its thresholds empirically on the development sample, without a held-out calibration step. For each candidate task, each class threshold is set to the lowest value at which at least an α fraction of the predictions accepted for that class agree with the oracle. Candidate tasks are assembled greedily w.r.t. minimizing token cost. At inference time, a document exits at the first stage whose proxy score exceeds the threshold.

4.2 Quality Estimation & Control

Let $\boldsymbol{\tau} = (\tau^-, \tau^+)$ denote the threshold configuration. We denote by $\Gamma_D(\boldsymbol{\tau})$ the fraction of documents resolved by the proxy, i.e., the oracle reduction rate (ORR), and by $Q_D(\boldsymbol{\tau})$ the accuracy of these proxy-resolved predictions. Since deferred documents are processed by the oracle, the overall accuracy is $\mathrm{Acc}_D(\boldsymbol{\tau}) = 1 - \Gamma_D(\boldsymbol{\tau})(1 - Q_D(\boldsymbol{\tau}))$. Threshold calibration estimates this quality using oracle-labeled samples and selects thresholds to minimize oracle deferral subject to $\mathrm{Acc}_D(\boldsymbol{\tau}) \geq \alpha$.

LOTUS [44] uses importance sampling mixed with uniform sampling and CLT-based normal approximation bounds. Let $\{(x_{i_j}, y_{i_j})\}_{j=1}^{m}$ denote the m oracle-labeled calibration draws, where x_{i_j} is sampled with replacement from D with probability p_{i_j}. For a threshold configuration $\boldsymbol{\tau}$, let $Z_j(\boldsymbol{\tau}) = \mathbb{I}[\widehat{y}_{i_j}(\boldsymbol{\tau}) \neq y_{i_j}]/n/p_{i_j}$ be the importance-weighted error indicator, where $\widehat{y}_{i_j}(\boldsymbol{\tau})$ is the final cascade prediction. Let $\overline{Z}(\boldsymbol{\tau})$ and $\widehat{\sigma}_Z(\boldsymbol{\tau})$ denote the sample mean and standard deviation of the m weighted error indicators, respectively. The lower accuracy bound is $L_{\mathrm{CLT}}(\boldsymbol{\tau}) = 1 - \overline{Z}(\boldsymbol{\tau}) - z_{1-\delta}\frac{\widehat{\sigma}_Z(\boldsymbol{\tau})}{\sqrt{m}}$, where $z_{1-\delta}$ is the $(1-\delta)$-quantile of the standard normal distribution.

BARGAIN [59] uses adaptive sampling and an anytime-valid betting-based test to certify proxy accuracy. For a threshold configuration $\boldsymbol{\tau}$, let $R(\boldsymbol{\tau}) \subseteq D$ be the set of documents resolved by the proxy, with $N = |R(\boldsymbol{\tau})|$. *BARGAIN* samples documents from this set without replacement. Let $X_j = \mathbb{I}[\widehat{y}_{i_j} = y_{i_j}]$ indicate whether the proxy prediction agrees with the oracle on the j-th sampled document. To test whether the accepted-region accuracy $Q_D(\boldsymbol{\tau})$ exceeds α, *BARGAIN* constructs a betting process of the form $K_t = \prod_{j=1}^{t}(1+\lambda_j(X_j - \mu_j))$, $\mu_j = \frac{N\alpha - \sum_{h=1}^{j-1} X_h}{N-j+1}$, where t is the number of observed labels, μ_j accounts for sampling without replacement under the null hypothesis $Q_D(\boldsymbol{\tau}) \leq \alpha$, and λ_j is a nonnegative betting fraction chosen from preceding observations and constrained to keep the process nonnegative. A threshold is certified when $K_t \geq 1/\delta$. Under the null hypothesis, the betting process satisfies $\mathbb{P}[\sup_t K_t \geq 1/\delta] \leq \delta$. Since $\mathrm{Acc}_D(\boldsymbol{\tau}) = 1 - \Gamma_D(\boldsymbol{\tau})(1 - Q_D(\boldsymbol{\tau}))$, certifying $Q_D(\boldsymbol{\tau}) \geq \alpha$ also guarantees the overall accuracy target.

ScaleDoc [60] uses stratified sampling to obtain oracle-labeled calibration data and reconstructs the positive and negative score distributions through interpolation and smoothing. Let $\widetilde{F}^+(u)$ and $\widetilde{F}^-(u)$ denote the estimated fractions of oracle-positive and oracle-negative documents, respectively, with proxy scores at most u. For $\boldsymbol{\tau} = (\tau^-, \tau^+)$, its accuracy adaptation estimates $\widehat{\mathrm{Acc}}_{\mathrm{SD}}(\boldsymbol{\tau}) = 1 - \widetilde{F}^+(\tau^-) - \big(\widetilde{F}^-(1) - \widetilde{F}^-(\tau^+)\big)$, where the two subtracted terms estimate false negatives and false positives resolved by the proxy. *ScaleDoc* selects the threshold pair that minimizes oracle deferral while meeting the estimated accuracy target. Its original theoretical analysis derives a Bernstein-based margin for an F1 target. Writing β for this original target, the margin is $\epsilon_{\mathrm{SD}} = (\sigma_W + (1-\beta)\sigma_P)\sqrt{\frac{4\ln(4/\delta)}{m}} + \frac{(8-6\beta)\ln(4/\delta)}{3m}$, where m is the calibration sample size, σ_W is the standard deviation of the weighted error indicator $W = (1-\beta/2)\mathbb{I}[y_i = 1, \widehat{s}_i < \tau^-] + (\beta/2)\mathbb{I}[y_i = 0, \widehat{s}_i > \tau^+]$, and σ_P is the standard deviation of $\mathbb{I}[y_i = 1]$.

Task Cascade [51] uses empirical accuracy on an oracle-labeled development set D_{dev} to determine class-specific thresholds at each proxy stage. For stage ℓ and predicted class $c \in \{0, 1\}$, let $S_{\ell,c}(\tau)$ be the subset of development documents predicted as class c with confidence at least τ. The empirical accuracy is $\widehat{Q}_{\ell,c}(\tau) = \frac{1}{|S_{\ell,c}(\tau)|}\sum_{x_i \in S_{\ell,c}(\tau)} \mathbb{I}[y_i = c]$. *Task Cascade* selects the smallest threshold τ for which $S_{\ell,c}(\tau)$ is nonempty and $\widehat{Q}_{\ell,c}(\tau) \geq \alpha$, where α is the target accuracy. Candidate stages are then assembled greedily to minimize inference cost. Unlike uncertainty-aware statistical bounds, this empirical criterion includes no uncertainty margin and provides no finite-sample guarantee in the base variant.

Table 2: Eight research questions in our evaluation framework.

Phase	Research Question	*LOTUS*	*BARGAIN*	*ScaleDoc*	*Task Cascade*
Proxy Construction	❶ How well do proxy scores distinguish positive and negative documents?	✘	✔✘	✔✘	✘
	❷ Does proxy deliver confidence correspond to practical accuracy?	✘	✔	✔✘	✘
	❸ How much oracle-reduction opportunity does a proxy provide, and how does the two-threshold cascade realize?	✘	✔✘	✔✘	✘
Proxy Cascade	❹ What Accuracy, ORR, and TSR does each system achieve?	✔	✔	✔	✔
	❺ What is the end-to-end token cost and execution latency?	✔	✔✘	✔	✔
	❻ How does performance vary with query selectivity?	✘	✔	✔	✘
	❼ How does performance vary across different oracle confidence score distributions?	✘	✔✘	✔✘	✘
	❽ What if replace surrogate models or calibration procedures across systems?	✔✘	✔	✔	✔✘

5 EVALUATION FRAMEWORK

We organize our study around eight research questions (RQ), spanning proxy quality, end-to-end performance, robustness, and component ablations. Table 2 summarizes these questions and their coverage in the original system studies. Here, RQ ❶ and RQ ❷ study the prediction quality and the beliefs of the surrogate model, respectively. RQ ❸ studies the upper limit of the oracle reduction rate of a proxy, which we define as a new posterior indicator *cascadablity*, that separates the opportunity supplied by a fixed proxy task from the ability of a practical system to exploit it. RQ ❹ and RQ ❺ study the overall effectiveness and efficiency of the four systems, respectively. In addition, RQ ❻ and RQ ❼ study the robustness of the four systems under two types of controlled variations of data distribution. Finally, RQ ❽ ablates the effect of cross-system components to study the interplay of surrogate model and score calibration. We use ✔ to denote that the question is directly evaluated, ✔✘ denotes that a similar question is explored, but the metrics, control variables, and factors are distinct, while ✘ denotes that the question is under-explored. The following subsections define the cascadeability diagnostic (§5.1) and the common evaluation protocol (§5.2).

5.1 Cascadeability & the Realization Gap

For a fixed proxy, cascadeability κ_α measures the maximum oracle reduction attainable at the accuracy target α. It is defined as the highest empirical ORR $\Gamma_D(\boldsymbol{\tau})$ achievable over all valid two-threshold configurations:

$$\kappa_\alpha = \max_{\boldsymbol{\tau}} \Gamma_D(\boldsymbol{\tau}) \quad \text{s.t.} \quad \mathrm{Acc}_D(\boldsymbol{\tau}) \geq \alpha. \tag{2}$$

Together, κ_α and achieved ORR characterize the oracle-reduction opportunity provided by the proxy and the extent to which the system's threshold calibration realizes this opportunity. To compute κ_α, we use a binary search over all proxy scores $\{\widehat{s}_i \mid x_i \in D\}$ to find the maximum range $[l^*, r^*]$ and the Accuracy of the cascade with $\tau^- = l^*, \tau^+ = r^*$ is no smaller than α, hence, $\kappa_\alpha = \Gamma_D(l^*, r^*)$. Since this computation requires complete oracle labels on D, κ_α serves as a retrospective diagnostic rather than a practical calibration method, and does not depend on the failure probability δ.

For a practical cascade with threshold configuration $\boldsymbol{\tau} = (\tau^-, \tau^+)$, we define its realization gap as $\kappa_\alpha - \Gamma_D(\boldsymbol{\tau})$. For runs satisfying the accuracy target, a smaller gap indicates that more of the available oracle-reduction opportunity is realized. A negative gap necessarily indicates a violation of the accuracy target. Varying α yields the cascadeability frontier $\alpha \mapsto \kappa_\alpha$, which is monotonically non-increasing as stricter accuracy targets reduce the feasible set of threshold configurations.

5.2 Evaluation Protocol

Model Settings. We use two LLM families, Qwen and Gemma, wherein Qwen 3.6 27B [48] and Gemma 4 12B [14] are used as the oracle models. The surrogate models of *LOTUS*, *BARGAIN*, and *Task Cascade* are the smaller LLMs in these two families, Qwen 3.5 9B [47] and Gemma 4 Effective 4B (E4B), respectively. Their input token monetary cost ($ per million tokens) is as follows:

Price ($)	Qwen 3.6 27B	Qwen 3.5 9B	Gemma 4 12B	Gemma 4 E4B
Input	0.29	0.08	0.10	0.02
Output	2.50	0.13	0.30	0.10

We use OpenAI o4-mini [42] for proxy task generation in *Task Cascade*. All the LLMs are locally deployed by SGLang [61] with chain-of-thought (CoT) reasoning disabled and a decoding temperature of 0. Following the original paper, we use NV-Embed-v2 [34] as the document embedding model for *ScaleDoc*.

Experimental Configurations. We set the calibration document set D' as 10% of the full document set D for all systems. In addition, a disjoint set of 10% of documents is randomly drawn from D as the training set for *ScaleDoc*'s surrogate model. The training hyperparameters of the lightweight surrogate model follow the original setting [60], including a 70/30 training-validation split, two training phases of 50 and 30 epochs, a learning rate of 10^{-3}, and a batch size of 128. Threshold calibration retains the default 64 histogram bins, a five-bin smoothing window, and 33 candidate thresholds. For *Task Cascade*, we follow the default three agentic proxy-generation iterations, five surrogate requests per iteration, and candidate document fraction of $\{0.1, 0.25, 0.5, 1.0\}$. By default, we set the accuracy target $\alpha = 0.95$ and the failure probability $\delta = 0.05$. When evaluating the Target Satisfaction Rate, we execute 100 independent runs per query on *LOTUS*, *BARGAIN*, and *ScaleDoc*. Unfortunately, *Task Cascade* spends 1.6-40.1 hours to process a single query. We execute 1 run per query for *Task Cascade*, and its TSR remains under-explored due to limited computational resources.

Environments. We implement the evaluation in Python. All experiments are conducted a Linux server with two AMD EPYC 9755

128-core CPUs, 503 GB of memory, and two NVIDIA H200 GPUs with 140GB of GPU memory.

6 TEST SUITE CONSTRUCTION

Our test suite comprises 10 dataset–predicate workloads collected from real-world document sets, complemented by our generated workload variants with controlled characteristics, providing evaluation conditions beyond the distributions represented by the original workloads. All variants reuse existing records and predicates without introducing additional source data.

6.1 Datasets and Predicates

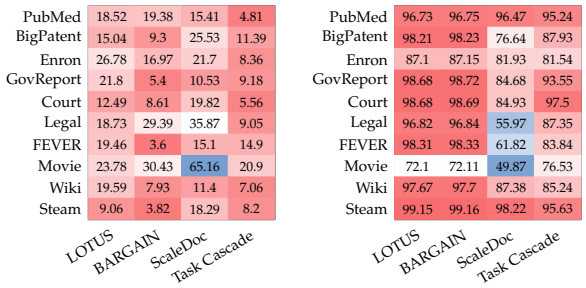
We evaluate the four semantic-filter systems on 254,369 records from ten datasets. Eight dataset–predicate workloads are adopted from prior work [44, 51, 59, 60], while we design new predicates for Wiki and Steam. Each dataset is paired with a natural-language binary predicate evaluated over a designated input field, requiring a `True`/`False` decision. The datasets cover scientific and technical text (PubMed [11] and BigPatent [52]), institutional and legal documents (Enron [32], GovReport [22], Court [15], and Legal [20]), fact verification and narrative understanding (FEVER [55] and Movie [24]), and user-generated content (Wiki [3, 8] and Steam [41]). Average predicate-input lengths range from 26 words for PubMed sentences to 23,095 words for Movie screenplays. Each dataset is paired with one natural-language binary predicate. We instantiate the predicate using the designated input field of each record and require a `True`/`False` decision. The predicates cover topical, relational, evidential, and pragmatic conditions rather than a single notion of document relevance. Table 3 summarizes the input fields, the number of documents $|D|$, average document lengths, predicate complexity, selectivity, and short summaries of predicates, along with dataset usage in the four original system studies. The symbols indicate whether the corresponding study uses the same dataset version (✔), a related dataset from the same family but with a different source or version (✔✘), or does not use the dataset (✘).

6.2 Controlled Workload Generation

Naturally occurring workloads reflect real-world data distributions but may not provide the variation in workload compositions needed for systematic evaluation. Therefore, we construct controlled workloads by selecting documents from collections in §6.1. For each oracle-model setting, we use the oracle's binary labels and predicted-label confidence to guide workload selection. The confidence of a record is defined as the token probability assigned to the oracle's final predicted label.

Label selectivity. We construct workloads with different proportions of oracle-positive records by selecting positive and negative records according to a specified label ratio. This allows us to evaluate how changes in class composition affect filtering quality, oracle reduction, and cascadeability under a common accuracy target.

Oracle confidence score. We construct four workload variants according to the oracle's predicted-label confidence, which reflects the oracle's uncertainty in making individual binary decisions, regardless of whether the predicted label is positive or negative. The `High` and `Low` variants select the 10,000 records with the highest and lowest confidence, respectively. The `Mixed` variant combines the 5,000 highest-confidence and 5,000 lowest-confidence records. The `Uniform` variant selects 3,000 records distributed across the confidence range. These variants provide workloads with different oracle-confidence characteristics, without changing the underlying documents or predicates.



(a) Brier Score (↓) (b) AUC-ROC (↑)

Figure 2: Brier Score & AUC-ROC of proxy scores (Qwen)

7 EXPERIMENTAL STUDY

RQ ❶: How well do proxy scores distinguish positive and negative documents?

Fig. 2 reports the ROC-AUC and Brier score of the proxy. A higher AUC indicates that proxy scores more consistently rank positive records above negative records. A lower Brier score indicates a smaller mean squared difference between the scores and binary oracle labels. *LOTUS* and *BARGAIN* achieve nearly identical AUC and generally rank positive records above negative records more consistently than *ScaleDoc* and *Task Cascade*. Although *ScaleDoc* achieves a high AUC on Steam, its ranking quality is considerably lower on Legal, FEVER, and Movie. *Task Cascade* exhibits more workload-dependent behavior: it surpasses both small-LLM proxies on Movie, but does not consistently retain this advantage across other workloads.

The Brier scores reveal a different aspect of proxy quality. *LOTUS* and *BARGAIN* exhibit substantially different Brier scores, despite their nearly identical AUC. *BARGAIN* generally achieves a lower Brier score, while *LOTUS* performs better on PubMed and Movie under both model settings. This difference partly reflects their score transformations: *LOTUS* applies quantile normalization, which changes numerical score values while largely preserving their ordering, whereas *BARGAIN* uses probability-derived scores. Given the low positive-label prevalence of this workload, a small squared error may imply strong separation between positive and negative records.

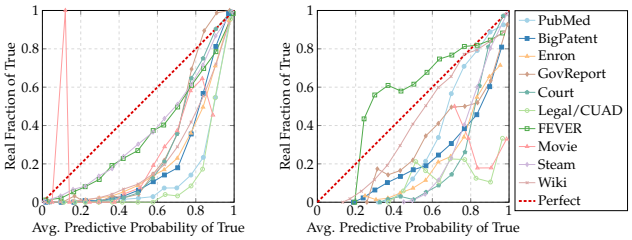
> **Finding 1:** Small LLM proxy with the original query as input generally performs the best in ranking and distinguishing positive and negative documents, consistently outperforming the lightweight model distilled from the oracle. The task-level proxy in *Task Cascade* performs well in specific workloads.

RQ ❷: Does proxy deliver confidence correspond to practical accuracy?

If a proxy score $\hat{s}_i$ is a calibrated estimate of the probability that a document receives a positive oracle label, then $\mathbb{P}(y_i = 1 \mid \hat{s}_i = s) = s$,

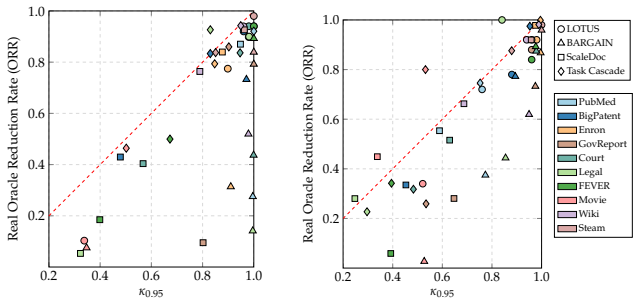
Table 3: Profile of 8 dataset-query workloads

Dataset	Predicate input	$\|D\|$	Avg. words	Predicate Difficulty	Gemma-12B Selectivity	Qwen-27B Selectivity	*LOTUS*	*BARGAIN*	*ScaleDoc*	*Task Cascade*	Predicate summary (True if …)
PubMed	Sentence	10,000	26	Medium	34.72%	19.20%	✗	✗	✓	✗	aims to evaluate a therapy's effect, efficacy, safety, or outcome
BigPatent	Abstract	10,000	107	Medium	37.57%	24.58%	✗	✗	✓	✗	claims to surpass or improve upon prior art
Enron	Email	9,997	324	Medium	6.53%	9.97%	✗	✗	✗	✓	was sent by a senior or high-ranking Enron employee
GovReport	Summary	10,000	500	Low	77.48%	85.29%	✗	✗	✓	✗	identifies a flaw or problem
Court	Opinion	16,902	2,816	High	71.36%	67.82%	✗	✓	✗	✓	reverses a lower-court ruling
Legal	Contract	489	8,176	High	47.24%	18.81%	✗	✗	✗	✓	contains a covenant not to sue or challenge IP rights
FEVER	Claim/evidence	15,981	343	High	81.05%	81.59%	✓	✗	✗	✓	contains at least one document that clearly supports the claim
Movie	Screenplay	1,000	23,095	High	17.50%	31.60%	✗	✓	✗	✗	protagonist makes a critical decision based on false information
Wiki	Comment	90,000	54	Medium	22.57%	16.54%	✗	✓	✗	✓	asks another editor a genuine question or requests clarification
Steam	Review	90,000	80	Low	42.72%	43.47%	✗	✓	✗	✓	expresses an overall positive opinion of the game



(a) Small-LLM proxy (b) *ScaleDoc*

Figure 3: Calibrations of proxy confidence (Qwen family)



(a) Qwen (b) Gemma

Figure 4: Real ORR vs. $\kappa_{0.95}$ (Qwen & Gemma families)

where y_i denotes the oracle label. Under this condition, accepting documents with $\hat{s}_i \geq \alpha$ yields $\mathbb{P}(y_i = 1 \mid \hat{s}_i \geq \alpha) = \mathbb{E}[\hat{s}_i \mid \hat{s}_i \geq \alpha] \geq \alpha$. A corresponding relation holds for the negative acceptance region with $\hat{s}_i \leq 1-\alpha$. However, the proxy scores of existing systems do not necessarily satisfy this calibration condition. Fig. 3 compares the mean proxy score in each bin with the corresponding fraction of positive documents. For the small LLM proxy, the observed positive fractions often fall below the predicted True probabilities across the score range. This suggests they tend to overestimate the predicted-label probabilities, reflecting a common problem with LLM token probabilities. In contrast, *ScaleDoc* exhibits more heterogeneous relationships across workloads. On FEVER, the observed positive fractions exceed the proxy scores across much of the intermediate score range, whereas on Movie and Legal, even high proxy scores correspond to relatively low positive fractions. This behavior is consistent with *ScaleDoc*'s scoring mechanism, which uses cosine similarities between contrastively trained document and query representations, optimized purely for class discrimination.

> **Finding 2:** Proxy confidence is not a reliable or transferable signal of agreement with the oracle. Both small LLM and lightweight models suffer from overconfidence in their predictions. Therefore, workload- and query-specific threshold recalibration is necessary for semantic filter systems to achieve the target accuracy.

RQ ❸: How much oracle-reduction opportunity does a proxy provide, and how does the two-threshold cascade realize?

Figs. 4 and 5 compare κ_α with realized ORR across systems and workloads. Low ORR can arise from two distinct bottlenecks: limited cascadeability or failure to exploit the available opportunity. At $\alpha = 0.95$, seven system–workload configurations exhibit low cascadeability ($\kappa_{0.95} < 0.5$), whereas five exhibit high cascadeability ($\kappa_{0.95} \geq 0.8$) but low realized ORR (< 0.5). For example, *ScaleDoc* exhibits both low cascadeability and low ORR on Movie, indicating that the constructed proxy offers limited opportunities for oracle reduction even with optimal thresholds. In contrast, *ScaleDoc* on GovReport and *BARGAIN* on Legal achieve low ORR despite substantially higher cascadeability.

Systems with comparable cascadeability can also differ substantially in how much of it they realize. *LOTUS* and *BARGAIN* differ in κ_α by less than 0.02 across all ten workloads, but *LOTUS* generally achieves at least 10% higher ORR on eight workloads, including five exceeding 40%. One possible contributing factor is threshold parameterization. The evaluated *BARGAIN* variant uses symmetric thresholds, whereas κ_α allows the two thresholds to vary independently. This restriction may exclude asymmetric acceptance regions that would otherwise satisfy the accuracy target while achieving higher ORR. Thus, even when proxies offer comparable oracle-reduction opportunities, differences in acceptance rules and threshold calibration can lead to substantially different realized ORR.

> **Finding 3:** Cascade effectiveness jointly depends on proxy quality and threshold calibration. Low ORR can arise from two separable bottlenecks: Proxies that offer little bypassing opportunity, and calibration strategies that fail to exploit the opportunity the proxy does offer. Both bottlenecks are observed among the four evaluated systems.

RQ ❹: What Accuracy, ORR, and TSR does each system achieve?

We evaluate four semantic filter systems across 10 workloads under two model families. Table 4 summarizes their performance at the default accuracy target $\alpha = 0.95$. We further evaluate *LOTUS*, *BARGAIN*, and *ScaleDoc* at accuracy targets ranging from 0.80 to 0.99. Fig. 6 reports target satisfaction across these targets, while

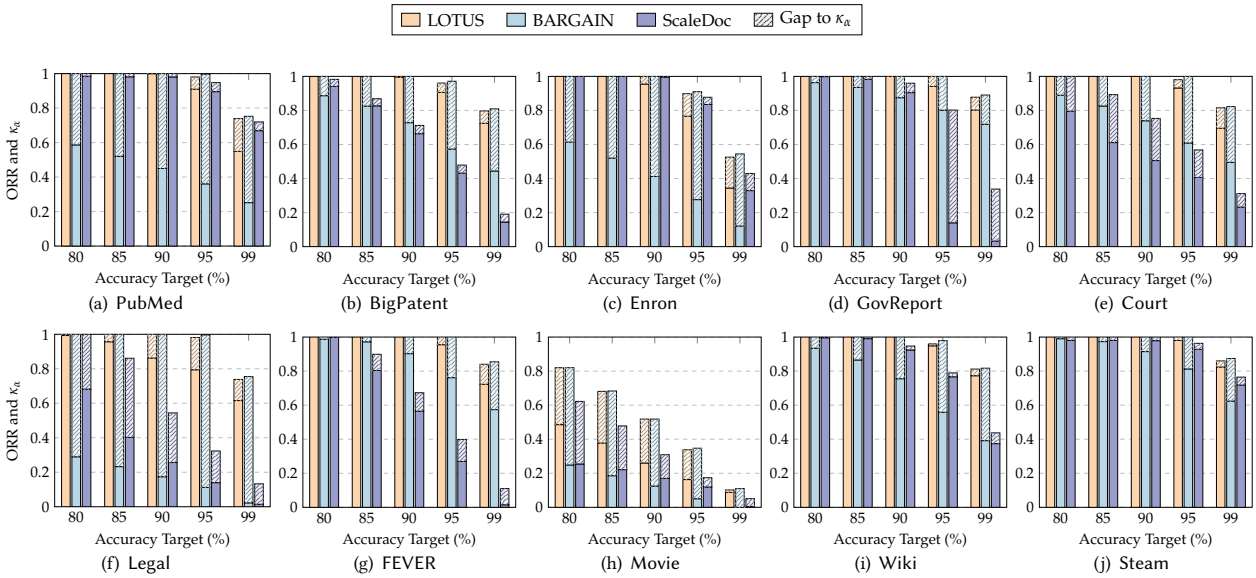


Figure 5: Oracle reduction rate and κ_α by varying accuracy target α (Qwen family)

Fig. 7 shows the distributions of achieved accuracy across repeated runs.

LOTUS achieves high oracle reduction but is less reliable on certain challenging workloads. Across the 10 workloads, it achieves the highest mean ORR among the four systems, while maintaining high overall TSRs of 98.2% and 97.1%. However, its failures are concentrated on Legal at the default target, and as the target increases, they become more frequent on both Legal and Movie. These failures are accompanied by different routing behaviors. On Legal, *LOTUS* continues to resolve a substantial fraction of records using the proxy despite frequent target violations. On Movie, its ORR decreases considerably, yet target violations persist, showing that increasing oracle fallback does not necessarily restore target satisfaction on challenging workloads. Both workloads contain long documents and require complex semantic judgments (Table 3).

BARGAIN maintains reliable target satisfaction, but its conservative behavior often compromises oracle reduction. At the default $\alpha = 0.95$, *BARGAIN* achieves the highest overall TSR among systems evaluated over repeated runs. Compared with *LOTUS*, its ORR is lower on every workload, yet its TSR is higher only on Legal and Movie. *Task Cascade* also achieves higher ORR than *BARGAIN* on nine workloads. This conservative behavior persists across accuracy targets. Even at relaxed targets ($\alpha = 0.80$ and 0.85), *BARGAIN* frequently achieves accuracy well above the required level while maintaining considerably lower ORR than *LOTUS*. As α improves, *BARGAIN* maintains high target at the expense of ORR, with TSR remaining at least 95% across the evaluated configurations, consistent with the specified failure probability $\delta = 0.05$.

ScaleDoc and *Task Cascade* perform well on specific workloads but exhibit inconsistent performance across datasets and model settings. At $\alpha = 0.95$, *ScaleDoc* achieves a TSR of at least 95% on only four of the ten workloads. While it combines high ORR with reliable target satisfaction on Steam, its high ORR on Enron is accompanied by frequent target violations. Conversely, on Legal, it achieves neither substantial oracle reduction nor reliable target satisfaction.

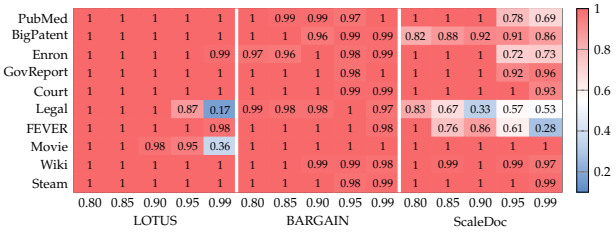


Figure 6: Target Satisfaction Rate (TSR) (Qwen family)

This difficulty persists at stricter accuracy targets, where increased oracle fallback fails to consistently restore target satisfaction on Legal. *Task Cascade* meets the accuracy target on eight of the ten workloads. It achieves high ORR while satisfying the target on PubMed and Wiki, but violates the target on BigPatent and Legal despite substantial oracle reduction.

> **Finding 4:** None of the evaluated systems consistently deliver both TSR and ORR across all workloads. *BARGAIN* achieves the most reliable Accuracy and TSR among the systems, with a little compromise of ORR. Semantic filter systems therefore favor an adaptive, workload-aware quality control design.

RQ ❺: What is the end-to-end token cost and latency?

Fig. 8 compares the token-based costs, while Fig. 9 reports the end-to-end latency of the four systems. The reference invokes the oracle directly on every document. Cost estimates only include model invocations for calibration and execution. *LOTUS* and *BARGAIN* show that higher oracle reduction does not necessarily translate into proportional cost or latency savings. Although *LOTUS* achieves higher ORR across all 10 workloads, it incurs higher token cost on four and longer latency on three. On GovReport and FEVER, *LOTUS* is both more expensive and slower than *BARGAIN* indicating

Table 4: Accuracy, ORR, and TSR of the four systems and proxy-switch ablations at $\alpha = 0.95$ (Qwen family). The best and second best results among the four systems, the improved vs. both components, and the unsatisfied are highlighted.

Dataset	*LOTUS*			*BARGAIN*			*Task Cascade*		*ScaleDoc*			*ScaleDoc* proxy & *LOTUS* calibration			*ScaleDoc* proxy & *BARGAIN* calibration		
	Acc.	ORR	TSR	Acc.	ORR	TSR	Acc.	ORR	Acc.	ORR	TSR	Acc.	ORR	TSR	Acc.	ORR	TSR
PubMed	97.26	90.95	100/100	98.45	36.48	97/100	97.42	92.08	95.53	90.06	78/100	97.11	85.40	100/100	97.60	20.96	100/100
BigPatent	97.16	90.58	100/100	99.02	57.56	99/100	94.75	83.35	95.65	43.77	91/100	97.53	30.46	100/100	97.41	8.60	99/100
Enron	97.28	76.59	100/100	98.00	27.97	98/100	96.06	79.35	95.38	85.17	72/100	97.32	67.11	100/100	97.01	7.22	97/100
GovReport	97.67	94.06	100/100	98.85	80.46	98/100	95.56	85.97	96.49	18.82	92/100	96.92	65.76	100/100	98.38	43.86	97/100
Court	97.15	93.21	100/100	98.78	61.07	99/100	96.11	83.64	97.22	40.66	100/100	96.76	46.03	100/100	98.22	30.81	100/100
Legal	97.17	81.54	87/100	98.39	11.19	100/100	92.84	92.64	95.11	18.51	57/100	95.94	26.11	74/100	99.98	0.02	100/100
FEVER	97.32	95.53	100/100	98.90	76.16	100/100	97.04	49.95	95.19	28.54	61/100	96.60	27.46	97/100	97.75	20.05	100/100
Movie	97.85	17.23	95/100	98.29	5.06	100/100	95.30	46.40	97.39	11.94	100/100	97.19	9.68	85/100	99.42	0.74	100/100
Wiki	96.16	94.87	100/100	98.91	56.17	99/100	95.00	94.24	95.45	76.70	99/100	96.08	72.86	100/100	98.36	44.71	97/100
Steam	96.11	97.99	100/100	98.80	81.67	98/100	95.25	83.73	96.06	92.80	100/100	96.17	92.86	100/100	98.51	37.86	99/100
Overall	97.11	83.26	982/1000	98.64	49.38	988/1000	95.53	79.13	95.95	50.70	850/1000	96.76	52.37	956/1000	98.26	21.48	989/1000

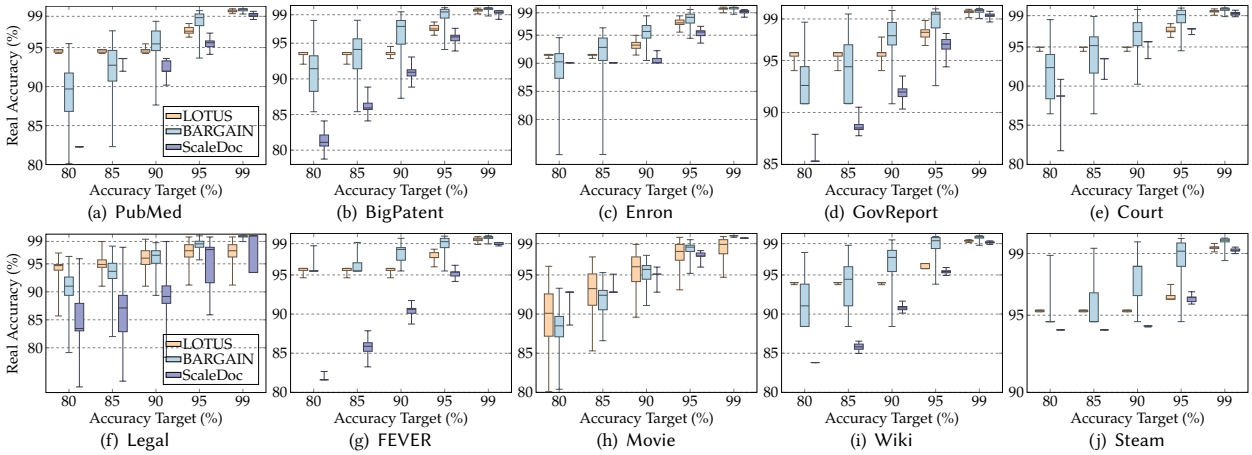


Figure 7: Accuracy by varying accuracy target α (Qwen family)

that proxy processing and calibration overhead can offset the savings from reduced oracle invocations. *ScaleDoc* reduces inference costs through its lightweight proxy, but its efficiency depends on oracle fallback and preprocessing overhead. It achieves lower token cost than oracle-only execution on all ten workloads, yet reduces end-to-end latency on only three. On Court and FEVER, frequent oracle fallback diminishes its cost advantage. Moreover, preprocessing accounts for a substantial share of its end-to-end latency, particularly on Legal, offsetting the benefits of inexpensive proxy inference. *Task Cascade* incurs substantial cascade-construction overhead that outweighs its inference savings. Despite reducing oracle calls during execution, its total reported cost and latency exceed those of oracle-only execution across all 10 workloads, with token costs 2.9-12.3× and end-to-end latencies 8.9-93.1 × those of the reference. Candidate task evaluation contributes substantially to its token cost, while preprocessing and calibration dominate its latency.

> **Finding 5:** ORR alone is not a comprehensive measure of efficiency. Proxy inference, calibration labeling, and cascade construction can offset or even outweigh the savings from oracle reduction. The lightweight surrogate of *ScaleDoc* is often token-efficient despite only moderate oracle reduction, but preprocessing can dominate its end-to-end latency on certain workloads. Whereas the cost of agentic multi-stage proxy construction even surpasses the cost of oracle invocations.

We evaluate the systems under workloads with varying label selectivity and the oracle LLMs' confidence. These workloads exhibit substantial variation in cascadeability κ_α, oracle reduction, target-hit rates, and filtering quality. Despite these differences, the overall findings remain consistent with our earlier observations: *BARGAIN* generally achieves higher accuracy at the expense of oracle reduction, while *ScaleDoc* exhibits less reliable target satisfaction on certain workloads.

RQ ❻: How does performance vary with query selectivity?

Fig. 10 shows that F1 generally increases with selectivity, with the largest differences between systems occurring when positive records are rare. *BARGAIN* maintains relatively high F1 at low selectivity, whereas *LOTUS* and *ScaleDoc* exhibit more pronounced declines. Despite these differences in F1, the achieved accuracy

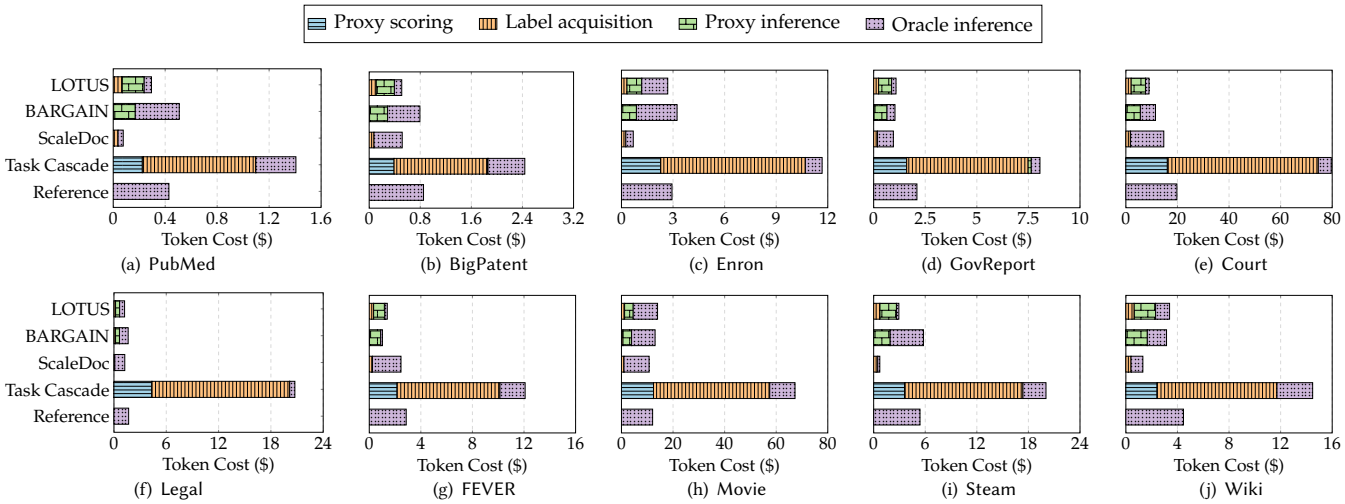


Figure 8: Token consumption ($) (Qwen family)

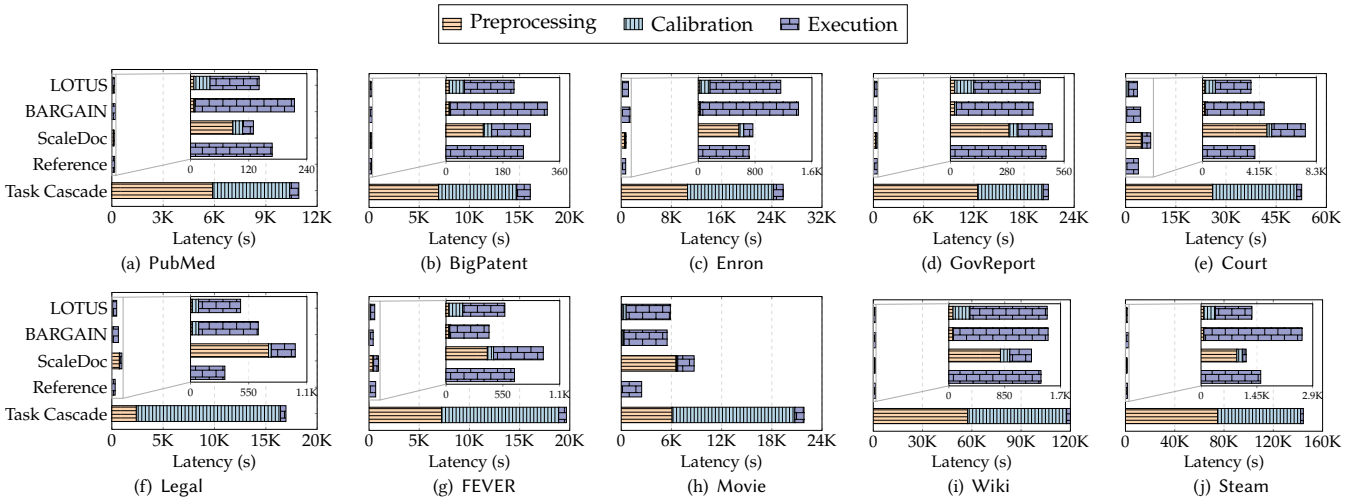


Figure 9: End-to-end latency (seconds) (Qwen family)

of *LOTUS* and *BARGAIN* remains relatively stable across selectivities. Moreover, *BARGAIN* consistently achieves lower ORR than *LOTUS*, suggesting that its higher F1 may partly result from more frequent oracle fallback rather than stronger proxy discrimination alone. Since F1 is sensitive to positive-class prevalence, meeting the same overall accuracy target does not necessarily yield comparable positive-class filtering quality, particularly on highly imbalanced workloads.

Selectivity also affects the oracle-reduction opportunities available under a fixed accuracy constraint. On Wiki, *ScaleDoc* exhibits substantially lower $\kappa_{0.95}$ near balanced selectivity than at more skewed label ratios, with realized ORR following a similar pattern. This dependence is less pronounced for some other systems, while on Steam, cascadeability remains relatively high across selectivities under both model settings. Part of this variation arises from the class composition of the workload: selectivity determines both how much each class contributes to overall ORR and how heavily its errors count toward the accuracy budget. When one class dominates, resolving a large fraction of that class can provide substantial oracle reduction, while errors in the minority class contribute less to overall accuracy loss. At more balanced ratios, errors from both classes carry greater weight in determining feasible acceptance regions.

> **Finding 6:** Selectivity changes which errors an accuracy target tolerates and how much opportunity a proxy can provide. Under class imbalance, minority-class errors weigh less in overall accuracy, so that high cascadeability or ORR may reflect uneven use of the accuracy budget instead of the strong discrimination of the system.

RQ ❼: How does performance vary across different oracle confidence score distributions?

We construct four workload distributions from Wiki and Steam using distinct selection rules based on oracle confidence: `Low`, `High`, `Mixed`, and `Uniform`. We construct `Low`, `High`, `Mixed`, and `Uniform`

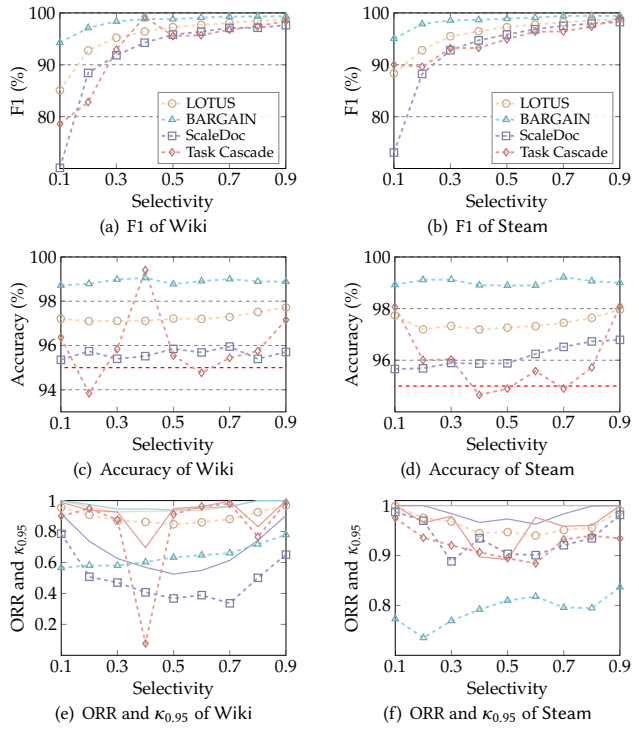


Figure 10: Sensitivity by varying selectivities (Qwen family)

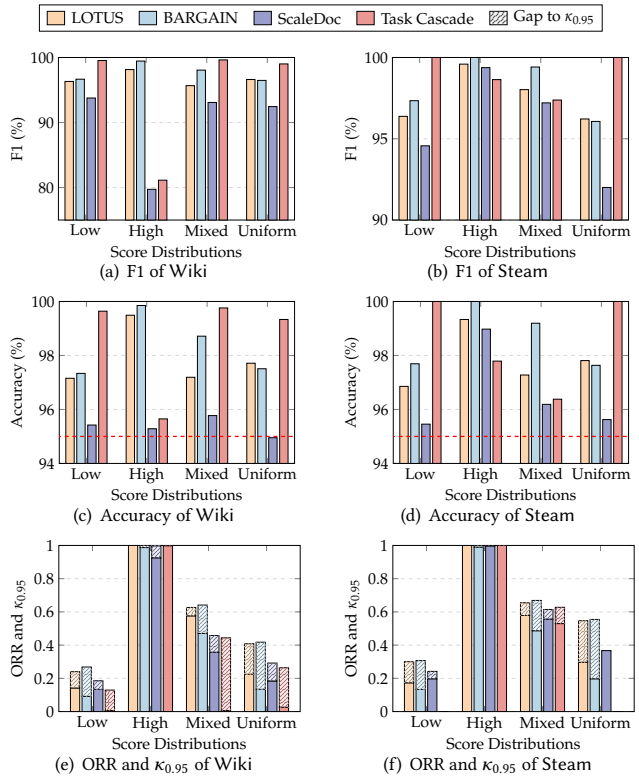


Figure 11: Sensitivity by varying oracle confidence score distributions (Qwen family)

workloads from Wiki and Steam using the oracle LLM's predicted-label confidence. Fig. 11 shows that `High` workloads have near-maximal $\kappa_{0.95}$ across all four systems, whereas `Low` workloads offer substantially less opportunity for oracle reduction. This pattern also holds for *ScaleDoc* and *Task Cascade*, whose proxy designs do not directly use the LLM's confidence scores.`Mixed` and `Uniform` workloads exhibit more varied cascadeability across systems and workloads. As a potentially useful signal, oracle uncertainty has yet to be explicitly exploited in system design. Notably, high cascadeability does not necessarily imply high F1: on `High` Wiki under Qwen, *ScaleDoc* and *Task Cascade* exhibit substantially lower F1 than *LOTUS* and *BARGAIN*, despite near-maximal $\kappa_{0.95}$.

Beyond the available opportunities, systems also differ in how effectively they realize cascadeability. On `High` workloads, realized ORR generally approaches $\kappa_{0.95}$, whereas larger realization gaps emerge under other confidence distributions. *LOTUS* generally realizes most of its available cascadeability, while *BARGAIN* exhibits larger gaps on `Uniform` than on `Mixed` workloads under both model settings. In contrast, *ScaleDoc* nearly reaches its cascadeability on `Uniform` Steam under Qwen, although this behavior does not persist under Gemma. *Task Cascade* also leaves substantial opportunities unrealized on `Mixed` Wiki, where its ORR remains near zero despite nontrivial cascadeability. These differences may partly arise from how systems sample records and estimate proxy quality. *BARGAIN* orders the documents according to the confidence and uses adaptive sampling to certify their quality. Compared with `Mixed` workloads, `Uniform` workloads include records beyond the confidence extremes, potentially introducing more heterogeneous error patterns that make expanded acceptance regions difficult to certify. *ScaleDoc* instead uses stratified sampling across score bins and reconstructs the score distributions through interpolation and smoothing, potentially contributing to its smaller realization gap on certain `Uniform` workloads.

> **Finding 7:** The underlying distribution of LLM confidence strongly influences the cascade opportunity for every proxy. Workloads with high LLM confidence are nearly fully cascadeable, whereas low LLM confidence leaves little cascade opportunity for existing proxies. Away from the high-confidence regime, realization gaps also widen in system-specific ways.

RQ ❸: What if we replace surrogate models or calibration procedures across systems?

We combine *ScaleDoc*'s proxy with the calibration and threshold selection procedures of *LOTUS* and *BARGAIN* to examine how these components perform across frameworks. Table 4 reports the results at an accuracy target of 95%, with 100 runs per configuration. Despite using identical underlying proxy scores, the three calibration procedures yield substantially different trade-offs between target satisfaction and oracle reduction. Compared with native *ScaleDoc*, *LOTUS* calibration substantially improves overall TSR while maintaining comparable ORR, whereas *BARGAIN* calibration achieves even higher TSR at the expense of considerably lower ORR. These differences are also workload-dependent: *LOTUS* calibration improves both TSR and ORR on GovReport under both model settings, but reduces both on Movie under Qwen. In contrast, both *LOTUS* and *BARGAIN* achieve lower ORR with *ScaleDoc*'s proxy than with

their native proxies, across all workloads and both model settings, without improving overall target satisfaction.

> **Finding 8:** Proxy construction and threshold calibration are not interchangeable modules, which should be designed and evaluated jointly.

8 RELATED WORK

LLM-powered Semantic Filtering. Semantic filtering has become a fundamental operator in LLM-powered data processing systems. Systems such as Palimpzest [39], DocETL [50], and SEMA [46] support natural-language predicates in declarative data processing pipelines, with optimizations including query rewriting, operator reordering, and efficient LLM execution. To reduce the cost of semantic filter evaluation, existing approaches either reduce the cost of per-record inference or avoid evaluating every record independently. The former can be achieved through selective model invocation or by replacing generative LLM inference with cheaper alternatives. FrugalGPT [5] employs model cascades that invoke more expensive LLMs only when needed. Other approaches use lightweight models to perform semantic judgments without generative inference. Chung et al. [6] train classifiers on document embeddings to replace selected LLM-based semantic operators, while JEVDB [57] employs Jev [56], a non-generative decision model that avoids autoregressive decoding. Beyond reducing the cost of per-record inference, other approaches leverage LLM judgments obtained from sampled records or previous queries to guide subsequent filtering. UQE [7] iteratively trains a classifier on LLM-labeled records to identify likely positive records for further LLM evaluation, aiming to maximize recall under a fixed inference budget. CSV [21] uses clustering, sampling, and voting to infer labels for semantically similar records, recursively refining ambiguous clusters. Building on CSV, Kim et al. [31] propose an adaptive two-phase method that switches to proxy-based filtering when clustering is insufficient. Beyond a single query, SemViews [18] reuses labels from previous semantic filters to improve the processing of new predicates, while using fresh oracle samples to certify result quality.

Approximate Query Processing. Approximate query processing (AQP) has long studied the trade-offs between query efficiency, result accuracy, and statistical guarantees [4]. Numerous AQP systems [1, 2, 16, 19, 25, 36, 43, 45] accelerate analytical queries over large relational datasets through sampling and precomputed data summaries, providing approximate query results with statistical error estimates. AQP has also been studied for queries involving expensive ML inference. Existing systems use inexpensive models to accelerate video filtering [27, 40] and support approximate aggregation through statistical estimation based on sampled oracle evaluations [26, 29, 49]. For approximate selection, SUPG [28] studies precision or recall guarantees under a fixed oracle budget, while Ding et al. [12] use proxy-based distance rankings to identify approximate query results, exploiting probabilistic error models or oracle sampling to meet precision or recall targets.

Benchmarks for Semantic Query Processing. Recent benchmarks evaluate LLM-powered semantic query processing from different perspectives. SemBench [33] constructs scenario-specific databases by combining existing manually annotated datasets across text, image, and audio modalities. Its queries combine semantic and relational operators, with ground-truth answers derived from the original dataset annotations. Bench-U [10] manually extracts structured attributes from unstructured documents across six domains to form relational tables. It generates analytical queries based on the resulting schemas, evaluates these queries using LLM-based systems over the original documents, and compares the returned answers with results obtained by executing equivalent queries on the relational tables. LROBench [53] designs natural-language queries over existing relational datasets, including BIRD [37], Magellan [9], NextiaJD [13], and SANTOS [30]. Three experts manually write SQL-like queries with LLM-enhanced operators and provide human-verified ground-truth results. The benchmark includes workloads involving single or multiple LLM-enhanced operators to evaluate individual implementations and their combinations. Same Query, Different Answer [17] repeatedly executes workloads from SemBench, Bench-U, and LROBench, showing that LLM output variability, even at temperature zero, can propagate through semantic query processing and affect the reliability of benchmark comparisons.

9 FUTURE RESEARCH DIRECTIONS

The benchmark covers single-predicate semantic filters over static document collections. Beyond this scope, we identify the following orthogonal directions as future research opportunities.

Filtering long documents exceed the context window. When a document exceeds the context window, even the oracle must read it in chunks, and the reference label may be obtained through an aggregation pipeline. *Task Cascade* has explored chunk-aware cascades, which exploit predicate structure and prefix sharing across reordered chunks. Open problems include effective segmentation and aggregation, certifying document-level accuracy from correlated chunk scores, and supporting asymmetric acceptance regions. The evaluation needs workloads that vary in length and evidence placement.

Reasoning-intensive predicates. Complicated predicates motivate oracles or proxies that use chain-of-thought (CoT) reasoning [58]. Introducing CoT poses two challenges for semantic filter processing. First, the token cost of CoT is charged by the relatively expensive output tokens. Systems favor a higher ORR with appropriate CoT compression techniques [35, 38, 54] for efficient processing. Second, the reasoning trace of small LLM surrogates can degenerate substantially, frequently becoming trapped in overthinking and self-reflection loops; hence, systems may require a mechanism for transferring the CoT from the oracle to small LLM surrogates.

Multimodal semantic filters. Images, audio, video, and scanned pages change both the cost structure and the choice of proxy. Lightweight per-query proxies have been proposed to accelerate video analytics [27, 40], and their design is similar to *ScaleDoc*. Whether a Multimodal LLM proxy also outperforms these lightweight proxies remains an open question.

Evolving collections and workload drift. In production-level applications, document collections evolve, and query predicates recur, while the evaluated systems calibrate thresholds once for a fixed collection and predicate. A guarantee certified on D does not necessarily hold for $D \cup \Delta D$, because data drift may invalidate calibrated thresholds. Since proxy scores are computed for every record, score

shifts are observable without oracle invocations and could trigger targeted recalibration. Systems entail a distribution shift detection and incremental recalibration mechanism to dynamically maintain the cascade framework with oracle label reuse.

Semantic Joins. *LOTUS* supports semantic join, which applies a predicate to document pairs. As the pair space is a Cartesian product, processing every pair with a surrogate model becomes prohibitively inefficient. Therefore, cascades need a candidate-generation stage, such as embedding-based blocking. This candidate-generation stage introduces challenges for a principled guarantee of quality targets. True positives may be lost during candidate-generation, which is invisible to threshold calibration unless the sampling covers the full pair space. However, true positives are typically sparse in the full pair space.

10 CONCLUSION

LLM-powered semantic filters are expensive when an oracle LLM evaluates every document sequentially. Proxy cascades reduce this cost only if an inexpensive surrogate model can resolve documents without violating a quality target. We present a unified benchmark that places four representative systems in a common proxy cascade framework and evaluates them under a common protocol. The test suite covers 10 real workloads across two LLM families, along with derivative synthetic workloads with controlled query selectivities and oracle confidence score distributions. We investigate 8 research questions, spanning proxy quality, end-to-end performance, robustness, and component ablations, and introduce 5 orthogonal future research directions.

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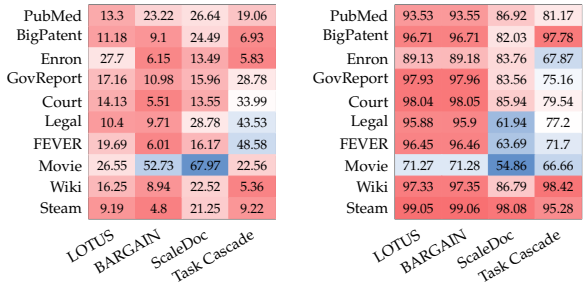
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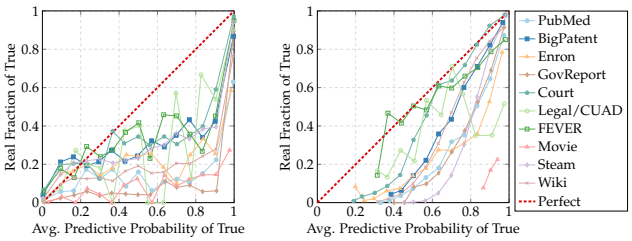
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A ADDITIONAL EXPERIMENTAL RESULTS



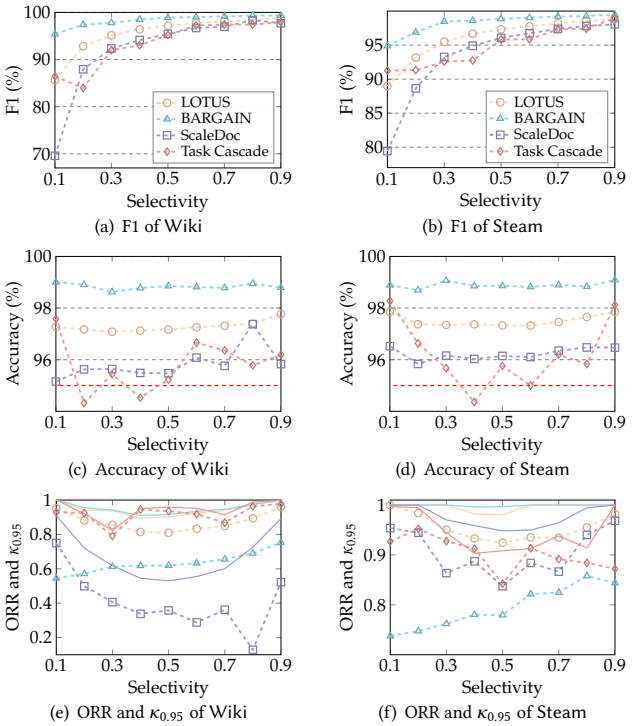
(a) Brier Score (↓) (b) AUC-ROC (↑)

Figure 12: Brier Score & AUC-ROC of proxy scores (Gemma)



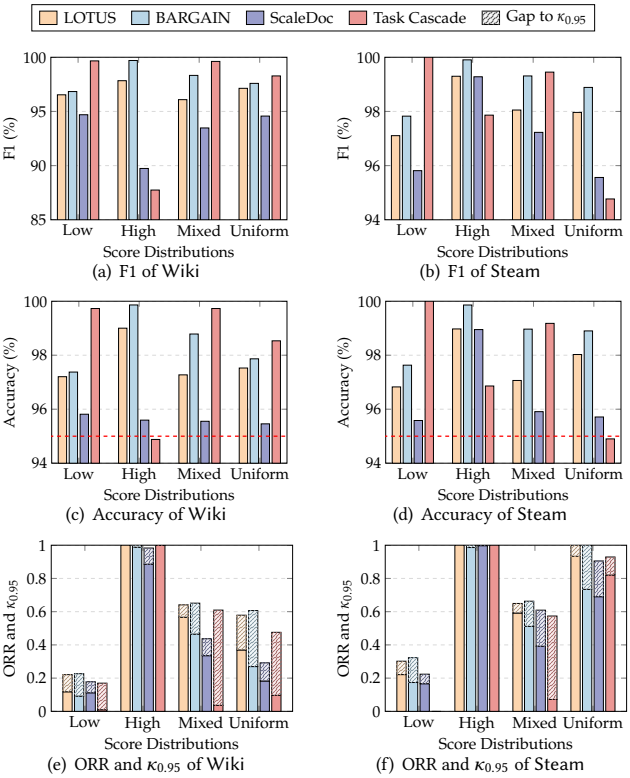
(a) Small-LLM proxy (b) *ScaleDoc*

Figure 13: Calibrations of proxy confidence (Gemma family)



(a) F1 of Wiki (b) F1 of Steam
(c) Accuracy of Wiki (d) Accuracy of Steam
(e) ORR and $\kappa_{0.95}$ of Wiki (f) ORR and $\kappa_{0.95}$ of Steam

Figure 14: Sensitivity by varying selectivities (Gemma family)



(a) F1 of Wiki (b) F1 of Steam
(c) Accuracy of Wiki (d) Accuracy of Steam
(e) ORR and $\kappa_{0.95}$ of Wiki (f) ORR and $\kappa_{0.95}$ of Steam

Figure 15: Sensitivity by varying oracle confidence score distributions (Gemma family)



Figure 16: Target Satisfaction Rate (TSR) (Gemma family)

Table 5: Accuracy, ORR, and TSR of the four systems and proxy-switch ablations at $\alpha = 0.95$ (Gemma family). The best and second best results among the four systems, the improved vs. both components, and the unsatisfied are highlighted.

Dataset	*LOTUS*			*BARGAIN*			*Task Cascade*		*ScaleDoc*			*ScaleDoc* proxy & *LOTUS* calibration			*ScaleDoc* proxy & *BARGAIN* calibration		
	Acc.	**ORR**	**TSR**	**Acc.**	**ORR**	**TSR**	**Acc.**	**ORR**	**Acc.**	**ORR**	**TSR**	**Acc.**	**ORR**	**TSR**	**Acc.**	**ORR**	**TSR**
PubMed	97.26	66.63	100/100	98.39	36.56	100/100	94.95	74.58	95.68	53.92	92/100	97.34	41.10	100/100	97.51	17.51	98/100
BigPatent	97.20	78.77	100/100	98.92	58.24	98/100	93.97	97.48	95.71	40.59	93/100	97.25	28.96	100/100	97.56	15.48	97/100
Enron	97.16	91.36	100/100	98.86	62.26	99/100	94.21	99.87	95.02	97.11	48/100	97.45	80.93	100/100	97.73	24.64	99/100
GovReport	97.57	88.54	100/100	98.98	67.32	100/100	98.17	25.90	95.93	42.62	94/100	96.94	48.88	100/100	98.14	34.89	98/100
Court	97.16	89.70	100/100	98.99	70.02	99/100	97.99	31.78	96.24	53.80	100/100	97.00	52.01	100/100	98.10	38.19	100/100
Legal	96.52	70.15	76/100	98.70	41.50	99/100	96.93	22.70	93.54	26.25	18/100	95.62	20.58	69/100	99.86	0.26	100/100
FEVER	97.10	89.08	100/100	98.68	66.11	99/100	98.70	34.18	96.35	6.58	100/100	96.62	26.52	100/100	97.67	19.78	99/100
Movie	97.75	26.04	95/100	98.65	2.60	100/100	87.20	80.00	95.12	31.61	60/100	97.11	15.87	86/100	99.72	0.33	100/100
Wiki	96.13	91.72	100/100	98.82	58.61	98/100	94.82	98.13	95.40	66.32	89/100	96.12	62.20	100/100	97.44	11.95	99/100
Steam	96.21	97.62	100/100	98.81	79.91	98/100	95.04	87.61	96.05	91.69	100/100	96.16	92.14	100/100	98.29	37.12	98/100
Overall	97.01	78.96	971/1000	98.78	54.31	990/1000	95.20	65.22	95.50	51.05	794/1000	96.76	46.92	955/1000	98.20	20.02	988/1000

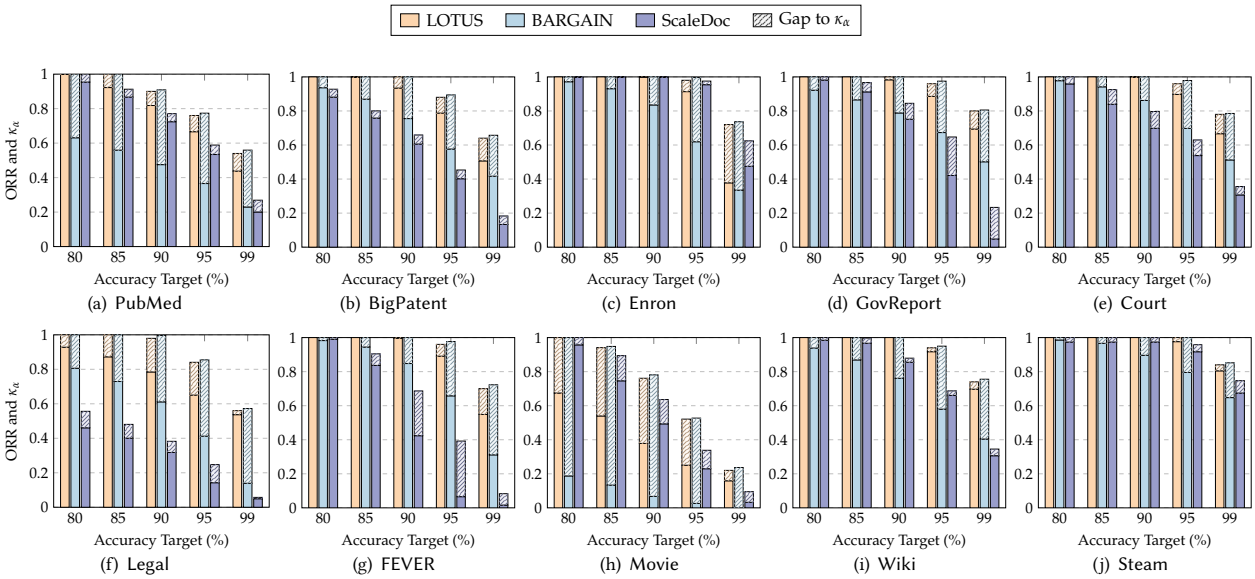


Figure 17: Oracle reduction rate and κ_α under by varying accuracy target α (Gemma family)

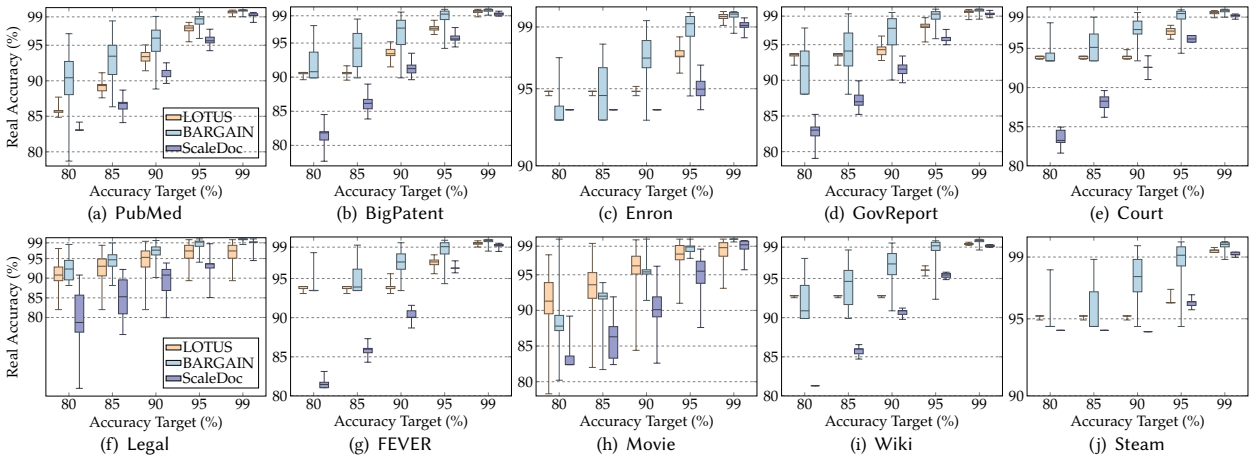


Figure 18: Accuracy varying accuracy target α (Gemma family)

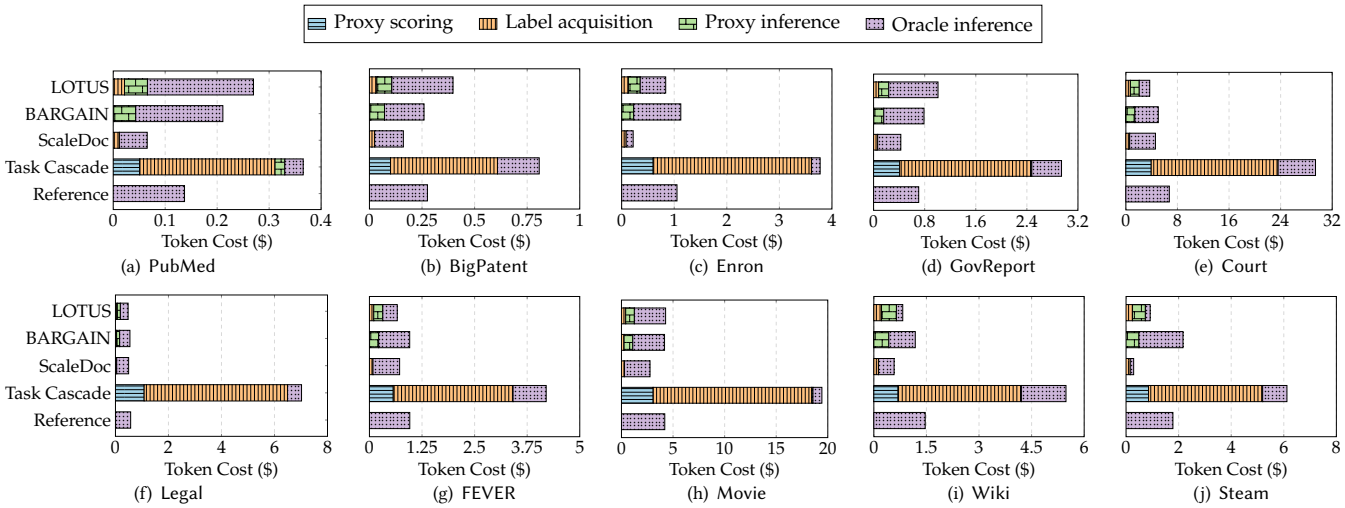


Figure 19: Token consumption ($) (Gemma oracle)

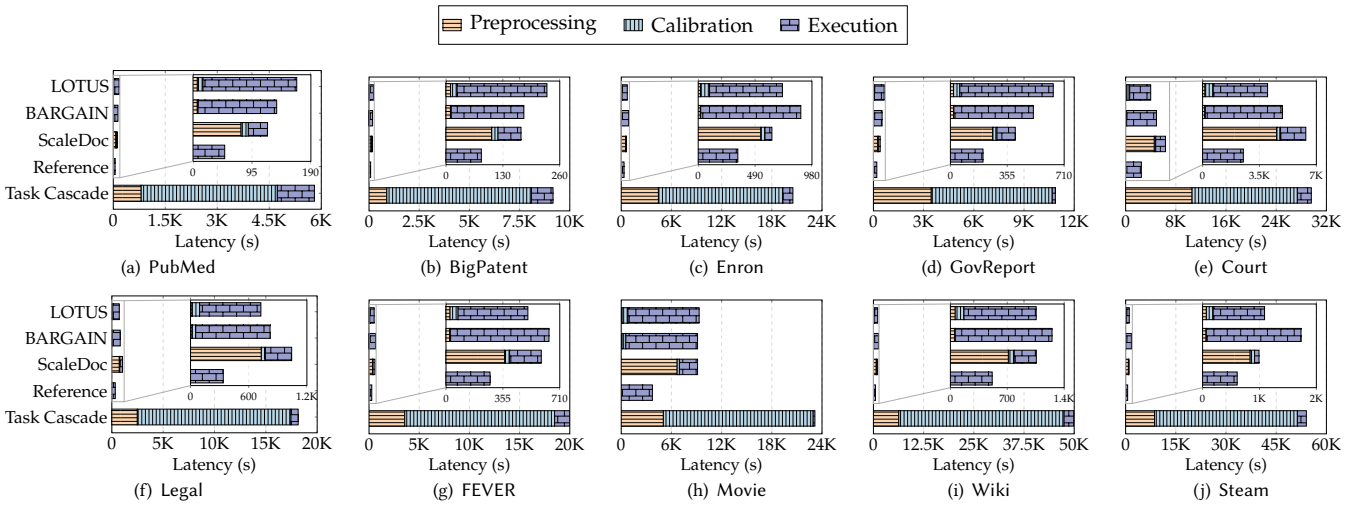


Figure 20: End-to-end latency (seconds) (Gemma family)